

BLUE LOTUS RESEARCH INSTITUTE

PhD-Level Investment Due Diligence — Healthcare Sector

Merck & Co., Inc.

[MRK] — Healthcare Due Diligence

Keytruda Monopoly — \$287B Market Cap · PD-1 Oncology Dominance · Keytruda \$25B+ Revenue · Pipeline Depth

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CLASSIFICATION: CONFIDENTIAL — INSTITUTIONAL RESEARCH

SECTION 1 — EXECUTIVE SUMMARY

Verdict: Strong Buy | Conviction: 9.1/10

Merck & Co., Inc. (MRK) — Healthcare Sector PhD Due Diligence. All prices and financials from BL3 MySQL (bluelotus3.historical_prices, ticker_fundamentals). Zero Claude training-data figures.
Verdict: Strong Buy. Conviction: 9.1/10.

Metric	Value
Price (BL3, 2026-08-07)	\$128.58
Market Cap	\$287B
P/E (TTM)	32.7x
P/B	6.3x
EPS (TTM)	\$7.28
Net Income (TTM)	\$18.0B
Dividend Yield	2.86%
52-Week High	\$123.33
52-Week Low	\$74.18

SECTION 2 — COMPANY OVERVIEW, BULL & BEAR CASE

Bull Case:

- Keytruda (pembrolizumab) is the world's best-selling drug: approved in 40+ cancer indications, \$25B+ annual revenue — PD-1 checkpoint inhibitor with 15+ year runway in combination regimens
- Keytruda subcutaneous formulation (SC-keytruda): new delivery format approved 2024 creates patent lifecycle extension of 5+ years beyond IV formulation expiry
- Winrevair (sotatercept): pulmonary arterial hypertension drug approved 2024 — \$5B+ peak revenue potential in an underserved market
- Vaccine franchise: Gardasil (\$8B+ revenue), globally dominant HPV vaccine with significant China expansion opportunity
- Pipeline optionality: MK-1654 (RSV), islatravir (HIV), MK-0616 (PCSK9 oral) — broad Phase 3 portfolio reduces single-drug concentration risk

Bear Case:

- Keytruda patent expiry: IV formulation patent expires 2028 — biosimilar entry (from 2028) will erode \$15-20B in revenue; SC formulation bridge is critical
- IRA price negotiation: Keytruda will enter Medicare negotiation — negotiations could reduce government reimbursement 30-50% for Medicare patients
- Gardasil China risk: China NIP (National Immunisation Programme) exclusion has constrained Gardasil volume; geopolitical risk to export licences
- Prometheus acquisition (\$10.8B): Tulisokibart (anti-TL1A for IBD) in Phase 3 — binary event risk for a high-price acquisition
- Concentration: Keytruda represents 35%+ of total MRK revenue — patent expiry creates a structural cliff even with portfolio diversification

SECTION 3 — FINANCIAL PERFORMANCE (BL3 MySQL)

Financial Metric (BL3 MySQL)	Value
P/E (TTM)	32.71x
Forward P/E	15.95x
P/B	6.25x
Net Income (TTM)	\$17.98B
EPS (TTM)	\$7.28
Market Cap	\$286.8B
Shares Outstanding	2,536M
Dividend Yield	2.86%
52-Week High	\$123.33
52-Week Low	\$74.18
Price (BL3, 2026-08-07)	\$128.58

SECTION 4 — PORTER'S FIVE FORCES (HEALTHCARE & PHARMA SECTOR)

Force	Intensity	Analysis
Rivalry	HIGH	Large-cap pharma (JNJ/MRK/PFE/ABBV/AMGN) compete on pipeline depth, patent coverage, and commercial scale. Generic manufacturers (Teva, Sandoz) and biosimilar developers erode branded margins upon patent expiry. Managed care (UNH/CVS) competes on network breadth and actuarial pricing.
New Entrants	LOW	Average drug development costs \$2-3B over 12-15 years with a 10-15% FDA approval probability. Hospital system integration, actuarial data, and physician relationships form additional managed care moats. Biotech startups enter as pipeline feeders rather than commercial competitors.
Supplier Power	MEDIUM	Active pharmaceutical ingredient (API) manufacturing is concentrated in China and India (70%+ of US API supply) — geopolitical disruption risk. Skilled oncologists and researchers command premium compensation. CMOs (Lonza, Catalent) are capacity-constrained for biologics manufacturing.
Buyer Power	HIGH	The three dominant pharmacy benefit managers (OptumRx/UNH, CVS Caremark, Express Scripts/Cigna) control formulary access for 270M+ Americans. Medicaid/Medicare price negotiation (IRA) now applies to 10 drugs initially. Hospital systems' GPO consolidation compresses medical device and specialty drug pricing.
Substitutes	HIGH	Patent cliffs expose \$250B+ in branded drug revenues to generic/biosimilar displacement 2024-2030. GLP-1 drugs (Ozempic, Mounjaro) are disrupting cardiovascular and obesity markets. CAR-T cell therapy and gene editing threaten haematology and oncology

		franchises. Direct primary care models pressure managed care margins at the margin.
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SECTION 5 — KEY RISKS

- Patent cliff wave 2024-2030: \$250B+ in branded pharma revenue exposed to generic/biosimilar entry — industry-wide earnings headwind
- IRA drug pricing: Medicare price negotiation now applies with expanding drug list — structural long-term reimbursement pressure on high-revenue drugs
- Clinical trial failure rates: Phase 3 failure rate is 30-40%; binary events create 20-50% stock dislocations for pipeline-dependent names
- GLP-1 disruption: Ozempic/Mounjaro reducing demand for diabetes devices, sleep apnoea, cardiovascular, and renal drugs — cross-sector disruption accelerating
- China trade risk: API manufacturing concentration in China creates supply chain vulnerability; US-China pharmaceutical decoupling is a multi-year risk
- Managed care regulatory pressure: CMS Medicare Advantage rate cuts, IRA drug pricing, and antitrust scrutiny of PBMs affect managed care and pharma margins

SECTION 6 — INVESTMENT THESIS & VERDICT

VERDICT: STRONG BUY | CONVICTION: 9.1/10

Merck & Co., Inc. (MRK) — Strong Buy with 9.1/10 conviction. BL3 data: Price \$128.58, MCap \$286.8B, NI \$17.98B, DivYld 2.86%. Healthcare sector remains a defensive compounding vehicle through the rate cycle.

Data: BL3 MySQL bluelotus3.ticker_fundamentals + historical_prices | CivilisationOS RAG (WSJ/FT/NIKKEI). Work Order: WO-DD-MEGA-HLTHC-20260814. Report Date: 14 August 2026. Zero training-data figures. AMP compliant.